

Linux Commands End to End Cheat Sheet

Command: whoami

- Description: Displays the current user account.
- Example: Running `whoami` in the terminal will show your username, like `john_doe`.

```
asif@asif-virtual-machine:~/Desktop$ whoami
asif
```

Command: ls

- Description: Lists files and directories in the current directory.
- Example: `ls` will display files and folders in your current directory, and `ls -l` provides detailed information.

```
asif@asif-virtual-machine:~/Desktop$ whoami
asif
asif@asif-virtual-machine:~/Desktop$ ls
asif@asif-virtual-machine:~/Desktop$ cd ~
asif@asif-virtual-machine:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
asif@asif-virtual-machine:~$
```

Command: ip a

- Description: Lists your IP address and network interfaces.
- Example: `ip a` show all network interfaces and their IP addresses.

```
asif@asif-virtual-machine:~$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | fou | help | ila | ioam | l2tp | link |
                  macsec | maddress | monitor | mptcp | mroute | mrule |
                  neighbor | neighbour | netconf | netns | nexthop | ntable |
                  ntbl | route | rule | sr | tap | tcpmetrics |
                  token | tunnel | tuntap | vrf | xfrm }
      OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                  -h[uman-readable] | -i[c] | -j[son] | -p[retty] |
                  -f[amily] { inet | inet6 | mpls | bridge | link } |
                  -4 | -6 | -M | -B | -O |
                  -l[oops] { maximum-addr-flush-attempts } | -b[rief] |
                  -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
                  -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
                  -c[olor]}
asif@asif-virtual-machine:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:3f:ba:5b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.84.128/24 brd 192.168.84.255 scope global dynamic noprefixroute ens33
        valid_lft 1688sec preferred_lft 1688sec
    inet6 fe80::6a42:20c6:cf41:15ee/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Command: cat

- Description: Reads the contents of text files.
- Example: `cat myfile.txt` will display the content of `myfile.txt`.

```
asif@asif-virtual-machine:~$ touch frame
asif@asif-virtual-machine:~$ ls
Desktop  Downloads  Music      Public  Templates
Documents  frame      Pictures  snap    Videos
asif@asif-virtual-machine:~$ vi frame
asif@asif-virtual-machine:~$ cat frame
hello hyd
:wq!

:wq!
```

•

Command: `passwd`

- Description: Used to change the password of the user account.
- Example: Typing `passwd` and following the prompts will allow you to change your password.

•

Command: `touch`

- Description: Creates empty files, mainly useful to test write ability.
- Example: `touch newfile.txt` creates an empty file named `newfile.txt`.

```
asif@asif-virtual-machine:~$ touch asif
asif@asif-virtual-machine:~$ echo $?
0
```

•

Command: `pwd`

- Description: Stands for Print Working Directory; shows which directory you are in.
- Example: Running `pwd` might return `/home/john_doe`.

```
asif@asif-virtual-machine:~$ pwd
/home/asif
```

•

Command: `man`

- Description: Accesses the system documentation for commands.
- Example: `man ls` will show the manual for the `ls` command.

```
asif@asif-virtual-machine:~$ man
What manual page do you want?
For example, try 'man man'.
```

•

Command: `apropos`

- Description: Searches commands related to a specific task based on description (equivalent to `man -k`).
- Example: `apropos "list files"` might return commands related to listing files.

```
asif@asif-virtual-machine:~$ apropos
apropos what?
```

•

Command: `pinfo`

- Description: Provides additional information, similar to `man`.
- Example: `pinfo ls` gives detailed information about the `ls` command.

Command: `tlldr`

- Description: Shows command line examples, a simplified version of man.
- Example: `tldr passwd` provides quick examples of how to use the `passwd`

```
asif@asif-virtual-machine:~$ tldr
tldr - Simplified and community-driven man pages

Usage: tldr [-v|--version]
          ((-u|--update) | [-p|--platform PLATFORM] COMMAND | (-a|--about))
CDROM client program

Available options:
  -h,--help          Show this help text
  -v,--version        Show version
  -u,--update         Update offline cache of tldr pages
  -p,--platform PLATFORM
                     Prioritize specific platform while searching. Valid
                     values include linux, osx, windows, sunos
  COMMAND            name of the command
  -a,--about          About this program
```

Essential File Management Tools Commands

Command: `ls`

- Example: `ls /home/user/Documents`
- Description: Lists files and directories within a specified path. This command is used to view the contents of directories.

```
to run a command as administrator (user "root"), use "sudo <command>".
see "man sudo_root" for details.

asif@asif-virtual-machine:~/Desktop$ whoami
asif
asif@asif-virtual-machine:~/Desktop$ ls
asif@asif-virtual-machine:~/Desktop$ cd ~
asif@asif-virtual-machine:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
asif@asif-virtual-machine:~$
```

Command: `cp`

- Example: `cp source.txt destination.txt`
- Description: Copies files from one location to another. This command is used to make duplicates of files.

```

asif@asif-virtual-machine:~$ cp frame newframe
asif@asif-virtual-machine:~$ ls -ltr
total 44
drwxr-xr-x 2 asif asif 4096 Aug  6 12:19 Videos
drwxr-xr-x 2 asif asif 4096 Aug  6 12:19 Templates
drwxr-xr-x 2 asif asif 4096 Aug  6 12:19 Public
drwxr-xr-x 2 asif asif 4096 Aug  6 12:19 Music
drwxr-xr-x 2 asif asif 4096 Aug  6 12:19 Downloads
drwxr-xr-x 2 asif asif 4096 Aug  6 12:19 Documents
drwxr-xr-x 2 asif asif 4096 Aug  6 12:19 Desktop
drwx----- 3 asif asif 4096 Aug  6 12:19 snap
drwxr-xr-x 3 asif asif 4096 Aug  6 15:11 Pictures
-rw-rw-r-- 1 asif asif  22 Aug  6 16:17 frame
-rw-rw-r-- 1 asif asif  22 Aug  6 17:16 newframe
asif@asif-virtual-machine:~$

```

Command: `mkdir`

- Example: `mkdir new_directory`
- Description: Creates a new directory. This command is used when you need to make a new folder.

```

asif@asif-virtual-machine:~$ mkdir shaik
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ ls
Desktop  Documents  Downloads  frame  Music  newframe  Pictures  Public  shaik  snap  Templates  Videos

```

Command: `rmdir`

- Example: `rmdir empty_directory`
- Description: Removes a directory, but only if it's empty. This command is used for deleting empty folders.

```

asif@asif-virtual-machine:~$ ls
Desktop  Documents  Downloads  frame  Music  newframe  Pictures  Public  shaik  snap  Templates  Videos
asif@asif-virtual-machine:~$ rmdir shaik
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$

```

Command: `mv`

- Example: `mv oldname.txt newname.txt`
- Description: Moves or renames files and directories. This command can change the location or the name of a file

```
asif@asif-virtual-machine:~$ mv newframe hello
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$
```

- **Command: `rm`**

- Example: `rm filename.txt`
- Description: Removes files or directories. This command deletes files or directories.
- Note: When combined with options `-r` (recursive) and `-f` (force), as in `rm -rf directory_name`, it becomes a powerful command that can delete directories and their contents without asking for confirmation.

```
asif@asif-virtual-machine:~$ rm frame
asif@asif-virtual-machine:~$ echo $?
0
```

-

Advanced File Management Tools Commands

ln: Used for creating hard and symbolic links.

- Example: `ln -s source_file symbolic_link` creates a symbolic link to `source_file`.
- Description: Links a file or directory to another location.

find: A utility to find files based on various properties.

- Example: `find /home -name "*.txt"` searches for text files in the `/home` directory.
- Description: Searches for files and directories in a given path.

```
asif@asif-virtual-machine:~$ find /home hello
/home
/home/asif
/home/asif/.local
/home/asif/.local/share
/home/asif/.local/share/gvfs-metadata
/home/asif/.local/share/gvfs-metadata/home
/home/asif/.local/share/gvfs-metadata/home-f72d8a0b.log
/home/asif/.local/share/gvfs-metadata/root
/home/asif/.local/share/gvfs-metadata/root-6c6e462f.log
/home/asif/.local/share/ibus-table
/home/asif/.local/share/sounds
/home/asif/.local/share/applications
/home/asif/.local/share/gnome-shell
/home/asif/.local/share/gnome-shell/gnome-overrides-migrated
/home/asif/.local/share/gnome-shell/application_state
/home/asif/.local/share/keyrings
/home/asif/.local/share/keyrings/login.keyring
/home/asif/.local/share/keyrings/user.keystore
/home/asif/.local/share/session_migration-ubuntu
/home/asif/.local/share/flatpak
/home/asif/.local/share/flatpak/db
/home/asif/.local/share/nautilus
/home/asif/.local/share/nautilus/scripts
/home/asif/.local/share/nautilus/tracker2-migration-complete
/home/asif/.local/share/nautilus/tags
/home/asif/.local/share/nautilus/tags/meta.db
/home/asif/.local/share/nautilus/tags/ontologies.gvdb
/home/asif/.local/share/recently-used.xbel
/home/asif/.local/share/icc
/home/asif/.local/share/gnome-settings-daemon
/home/asif/.local/share/gnome-settings-daemon/input-sources-converted
/home/asif/.local/share/evolution
```

locate: Used for quickly searching the database of system's contents.

- Example: `locate filename` finds the location of `filename`.
- Description: Fast file searching, requires running `updatedb` for an updated database.

```

asif@asif-virtual-machine:~$ locate hello
boot/grub/i386-pc/hello.mod
boot/grub/x86_64-efi/hello.mod
/home/asif/hello
/snap/core22/1612/usr/lib/python3.10/_phello_.foo.py
/snap/core22/1612/usr/lib/python3.10/_pycache_/_phello_.foo.cpython-310.pyc
/snap/gnome-42-2204/176/usr/lib/python3.10/_phello_.foo.py
/snap/gnome-42-2204/176/usr/lib/x86_64-linux-gnu/peas-demo/plugins/helloworld
/snap/gnome-42-2204/176/usr/lib/x86_64-linux-gnu/peas-demo/plugins/pythonhello
/snap/gnome-42-2204/176/usr/lib/x86_64-linux-gnu/peas-demo/plugins/helloworld/helloworld.plugin
/snap/gnome-42-2204/176/usr/lib/x86_64-linux-gnu/peas-demo/plugins/helloworld/libhelloworld.so
/snap/gnome-42-2204/176/usr/lib/x86_64-linux-gnu/peas-demo/plugins/pythonhello/pythonhello.plugin
/snap/gnome-42-2204/176/usr/lib/x86_64-linux-gnu/peas-demo/plugins/pythonhello/pythonhello.py
usr/lib/grub/i386-pc/hello.mod
usr/lib/grub/x86_64-efi/hello.mod
usr/lib/python3.10/_phello_.foo.py
usr/lib/python3.10/_pycache_/_phello_.foo.cpython-310.pyc
usr/share/doc/python3-uno/demo/hello_world_comp.py
usr/share/locale-langpack/en@quot/LC_MESSAGES/hello.mo
usr/share/locale-langpack/en_AU/LC_MESSAGES/hello.mo
usr/share/locale-langpack/en_CA/LC_MESSAGES/hello.mo
usr/share/locale-langpack/en_GB/LC_MESSAGES/hello.mo
asif@asif-virtual-machine:~$ echo $?
0

```

which: Finds the location of executables in the PATH.

- Example: `which python` shows the path of the Python executable.
- Description: Useful for finding executable files in system's PATH.

tar: Archives multiple files into a single file.

- Example: `tar -cvf archive.tar /folder` creates an archive of a folder.
- Description: Combines multiple files into a tarball.

gzip: Compresses files.

- Example: `gzip file.txt` compresses `file.txt`.
- Description: Reduces the size of files using compression.

```

asif@asif-virtual-machine:~$ gzip hello
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$

```

bzip2: A file compression utility.

- Example: `bzip2 file.txt` compresses `file.txt`.
- Description: Offers better compression ratios than `gzip` at the cost of speed.

```
asif@asif-virtual-machine:~$ bzip2 hello.gz
asif@asif-virtual-machine:~$ echo $?
```

- 0

xz: High compression utility.

- Example: `xz file.txt` compresses `file.txt`.
- Description: Provides the best compression ratio.

```
asif@asif-virtual-machine:~$ xz hello.gz.bz2
asif@asif-virtual-machine:~$ echo $?
```

```
0
```

- asif@asif-virtual-machine:~\$

zip: Used for packaging and compressing (archiving) files.

- Example: `zip archive.zip file1 file2` creates an archive with `file1` and `file2`.
- Description: Commonly used for creating `.zip` files.

```
asif@asif-virtual-machine:~$ zip archive.zip asif lenovo
adding: asif (stored 0%)
adding: lenovo (stored 0%)
asif@asif-virtual-machine:~$ echo $?
```

```
0
```

file: Determines the type of a file.

- Example: `file image.jpg` displays the type of `image.jpg`.
- Description: Useful for identifying file formats.

mount: Attaches a filesystem to a specified directory.

- Example: `mount /dev/sdb1 /mnt` mounts the device `sdb1` to `/mnt`.

- ```
/usr/bin/cp --recursive -mount $mnt
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
devpts on /dev type devtmpfs (rw,nosuid,relatime,size=3998064k,nr_inodes=999671,nodev=755,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=800)
misc on /run type tmpfs (rw,nosuid,nodev,noexec,relatime,size=883396k,nodev=755,inode64)
securityfs on / type ext4 (rw,relatime,errors=remount-ro)
configs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64)
tmpfs on /run/lock type tmpfs (rw,nosuid,nodev,noexec,relatime,size=512K,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot)
hugetlb on /sys/hugepages type hugetlbfs (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=750)
systemd-1 on /proc/sys/fs/binfmt_misc type auto (rw,relatime,fd=29,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=12300)
nfsd4 on /dev/nfsd4 type nfsd4 (rw,relatime)
queue on /dev/queue type queue (rw,nosuid,nodev,noexec,relatime)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
tracing on /sys/kernel/tracing type tracingfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
rpc on /run/rpc_pipefs type rpc_pipefs (rw,nosuid,nodev,noexec,relatime,size=700)
vmware-vmblock on /run/vmblock-fuse type fuse.vmware-vmblock (rw,relatime,user-id=0,group-id=0,default_permissions,allow_other)
snapd/snapshots/barf-snap on /snap/barf/5 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
snapd/snapshots/gnome-4284-snap on /snap/gnome-4284/1012 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
snapd/snapshots/firefox-4284-snap on /snap/firefox/4848 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
snapd/snapshots/gnome-4284-176-snap on /snap/gnome-4284-176 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
snapd/snapshots/gnome-4284-176-snap on /snap/gnome-4284-176 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
snapd/snapshots/store-1113-snap on /snap/store-1113 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
snapd/snapshots/snap-store-1113-snap on /snap/snap-store/1113 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
snapd/snapshots/snap-store-21759-snap on /snap/snap-store/21759 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
snapd/snapshots/snap-desktop-integration-178-snap on /snap/desktop-integration/178 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide)
dsad2 on /boot/efi type vfat (rw,relatime,fmask=0077,dmask=0077,codepage=437,iocharset=utf8,shortname=mixed,errors=remount-ro)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,noexec,relatime,size=8392k,nodev=755,inode64,inode64)
portal on /run/user/1000/doc type fuse.portal (rw,nosuid,nodev,noexec,relatime,user-id=1000,group-id=1000)
tmpfs on /run/udev/mqueue type tmpfs (rw,nosuid,nodev,noexec,relatime,size=883396k,nodev=755,inode64)
msdosfs on /run/media/msd/snap-desktop-integration type msdosfs (rw,nosuid,nodev,noexec,relatime)
gvfsd-fuse on /run/user/1000/gvfs type fuse.gvfsd-fuse (rw,nosuid,nodev,relatime,user-id=1000,group-id=1000)
devfsd on /media/asf/ubuntu 22.04.5 LTS amd64 type iso9660 (rw,nosuid,nodev,relatime,nojoliet,checkas,mapain,blocksizet=2048,uid=1000,gid=1000,mode=500,rmode=400,tocachert=ut8,whelpar=udisks2)
/dev/sr0 on /media/asf/CROM type iso9660 (rw,nosuid,nodev,relatime,nojoliet,checkas,mapain,blocksizet=2048,uid=1000,gid=1000,mode=500,rmode=400,tocachert=ut8,whelpar=udisks2)
#dfsgaffix-virtual-machine # echo $?
```

- Example: `df -h` shows disk space in human-readable format.
- Description: Useful for monitoring disk space usage.

```
asif@asif-virtual-machine:~$ df
Filesystem 1K-blocks Used Available Use% Mounted on
tmpfs 808396 1960 806436 1% /run
/dev/sda3 19946096 12126884 6780672 65% /
tmpfs 4041964 0 4041964 0% /dev/shm
+
+ Floppy Disk 5120 4 5116 1% /run/lock
+ , 524252 6232 518020 2% /boot/efi
tmpfs 808392 108 808284 1% /run/user/1000
/dev/sr1 4651082 4651082 0 100% /media/asif/Ubuntu 22.04.5 LTS amd64
/dev/sr0 158436 158436 0 100% /media/asif/CDROM
asif@asif-virtual-machine:~$ echo $?
0
```

- **Example:** `findmnt -t ext4` lists all ext4 type mounts.
- **Description:** Displays a tree-like structure of mounts.
- 

- Example: `lsblk` lists all block devices.

- Description: Useful for identifying storage devices and their partitions.

```
asif@asif-virtual-machine:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
fd0 2:0 1 4K 0 disk
loop0 7:0 0 4K 1 loop /snap/bare/5
loop1 7:1 0 74.3M 1 loop /snap/core22/1612
loop2 7:2 0 2M 1 loop /snap/firefox/4848
loop3 7:3 0 309.1M 1 loop /snap/gnome-42-2204/176
loop4 7:4 0 91.7M 1 loop /snap/gtk-common-themes/1535
loop5 7:5 0 12.9M 1 loop /snap/snap-store/1113
loop6 7:6 0 38.8M 1 loop /snap/snapd/21759
loop7 7:7 0 500K 1 loop /snap/snapd-desktop-integration/178
sda 8:0 0 20G 0 disk
├─sda1 8:1 0 1M 0 part
├─sda2 8:2 0 513M 0 part /boot/efi
└─sda3 8:3 0 19.5G 0 part /
sr0 11:0 1 154.7M 0 rom /media/asif/CDROM
sr1 11:1 1 4.4G 0 rom /media/asif/Ubuntu 22.04.5 LTS amd64
asif@asif-virtual-machine:~$ echo $?
0
```